MLH HACK DAY · TECHNICAL PROJECT

CampusFix: Smarter Campus Issue Management

Report campus issues. Prioritize what matters. Reward community action.

Gemma 4 · [INSERT TEAM MEMBERS] · [INSERT COLLEGE] · [INSERT DEPARTMENT]



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THE PROBLEM

Campus problems need a visible path to action

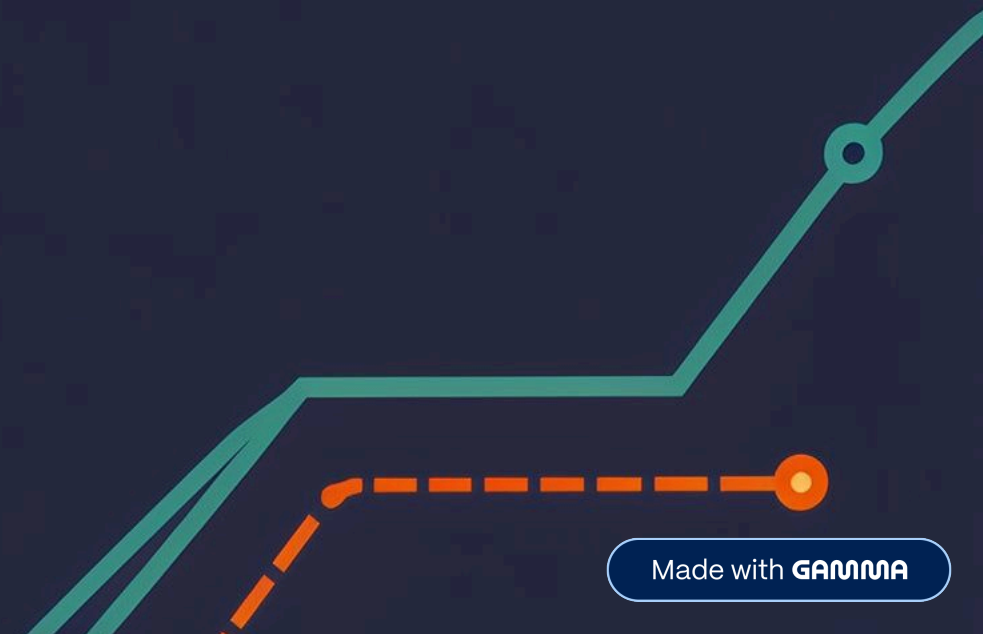
Issues can go unreported, unnoticed, or unresolved.

Safety-critical problems need attention before routine maintenance.

Repeated confirmations reveal community impact.

Students need transparent ways to contribute and receive recognition.

When the path is invisible, impact is invisible too.



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THE PARTICIPATION LOOP

Report, prioritize, participate

Each action creates a clearer next step.

01

Report

Submit a photo and description.

02

Issue board

Browse, confirm, and resolve legitimate fixes.

03

Leaderboard

Earn points through useful reporting and community action.

REPORT → AI-ASSISTED CLASSIFICATION → SAFETY VALIDATION → RANKED BOARD → COMMUNITY ACTION → POINTS

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SYSTEM BOUNDARIES

Gemma 4 judges meaning; Python controls the system

INPUT

Student or staff → Gradio

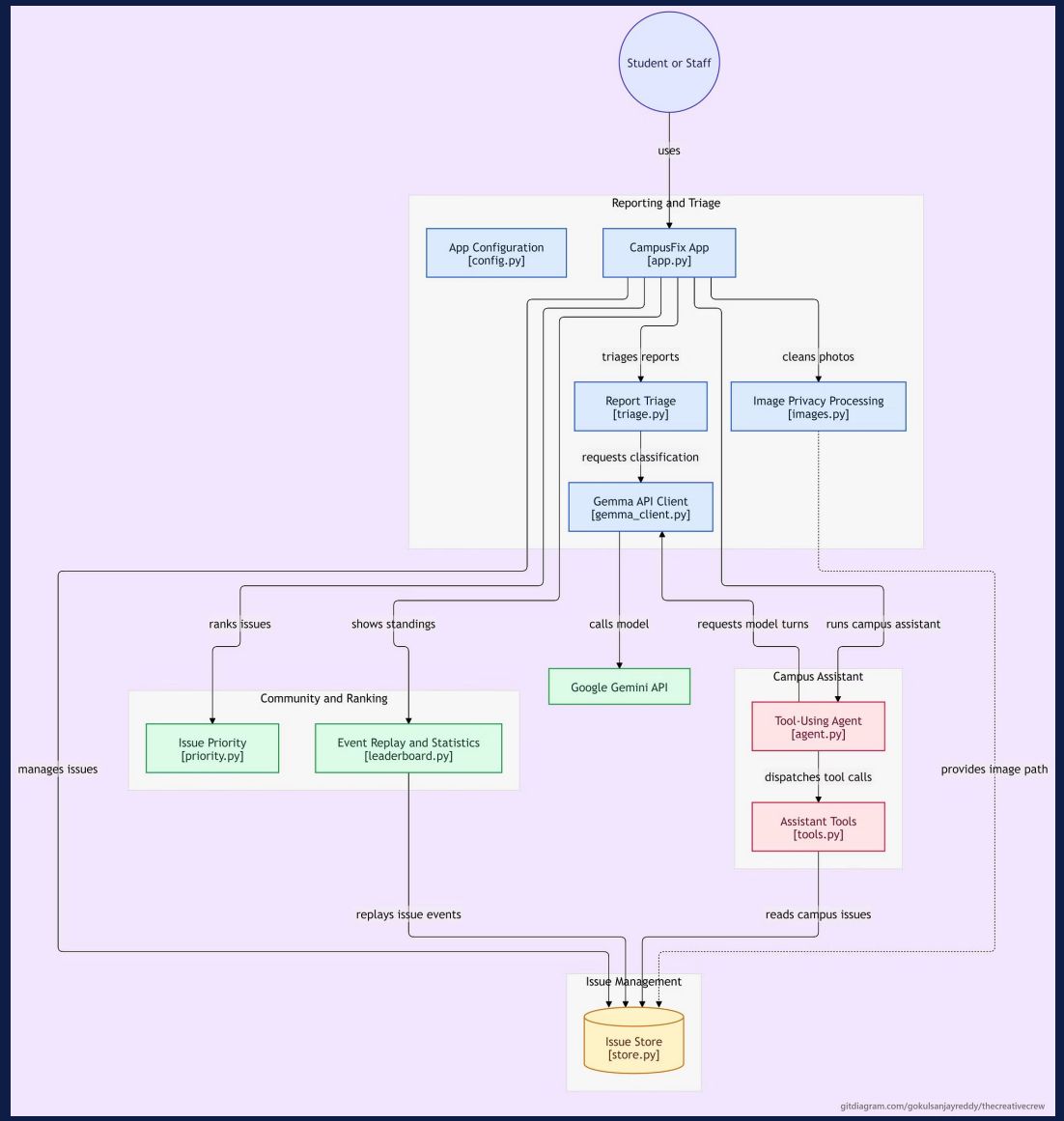
CONTROL

Image validation → Python safety floor

OUTPUT

JSON store → ranked issue board

Python modules: images.py, triage.py, priority.py, leaderboard.py, store.py. Leaderboard replays stored events; board actions and points do not call Gemma 4.



Gemma 4 via Google Gemini API · model gemma-4-26b-a4b-it

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CONTROLLED WORKFLOW

From report to action in six controlled steps

Submit

Description and optional photo.

Clean

Validate type and size; strip EXIF; resize to max 1280px; save as JPEG.

Classify

Gemma 4 returns category, severity, safety risk, location hint, and reason.

Validate

Python parses JSON and applies rules; safety can raise severity, never lower it.

Prioritize

Score severity, age, and confirmations.

Act and reward

Store events, show the board, and replay events for points.

PRIORITY SCORE

$$PriorityScore = SeverityBase + \min(\text{Age in Days}, 40) \times 1.5 + \min(\text{Confirmations} \times 2, 10)$$

Bases: Critical 100 · High 60 · Medium 30 · Low 10

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FAIRNESS BY DESIGN

Four engineering choices make the system fairer

01 COST

One AI call per report

Routine board and leaderboard actions use zero Gemma calls.

02 SAFETY

Deterministic safety rules

Sparking wires, smoke, and gas leaks can raise severity to high or critical.

03 PRIORITY

Aging-aware priority

Aging caps at 40 days; confirmations cap at 10 points; older timestamps break ties.

04 ACCOUNTABILITY

Auditable leaderboard

Points come from replayed events; limits and resolution rules reduce abuse.

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PRODUCT WALKTHROUGH

See CampusFix in action

Illustrative UI — replace with actual screenshot

A · REPORT

Sparking electrical wire near the library entrance.

Photo upload

Optional photo

Text submission · illustrative example

B · ISSUE BOARD

Electrical safety issue

Severity	High
Age · Confirmations	2 days · 7
Priority	Ranked

+1 confirm · resolve · reopen, if implemented

C · LEADERBOARD

Community action

Reporting	logged
Confirmations	logged
Resolving	logged

Points follow logged event rules.

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RELIABILITY

Reliability protections support practical impact

PRIVACY

Image privacy

EXIF metadata is removed before saving images.

SAFETY

Safe triage

Deterministic rules can raise hazardous reports to high or critical severity.

FAULT TOLERANCE

Fault tolerance

API failure falls back to category Other, severity medium, and needs_review=true.

STORAGE

Reliable storage

Thread locks, temporary-file writes, atomic replacement, and corrupt-file backup protect issue data.

Intended impact: a clearer queue, more transparent prioritization, and a structured way for students to contribute.

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NEXT HORIZON

Future scope: production, operations, evaluation

These are proposed improvements, not current features.

01

Production deployment

Authenticated accounts, role-based staff access, secure hosted storage.

02

Smarter operations

Notifications, assignment, escalation, and resolution-time tracking.

03

Improved evaluation

Test classification quality, safety-rule coverage, API failure handling, and leaderboard abuse resistance.

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CAMPUSFIX · THE TAKEAWAY

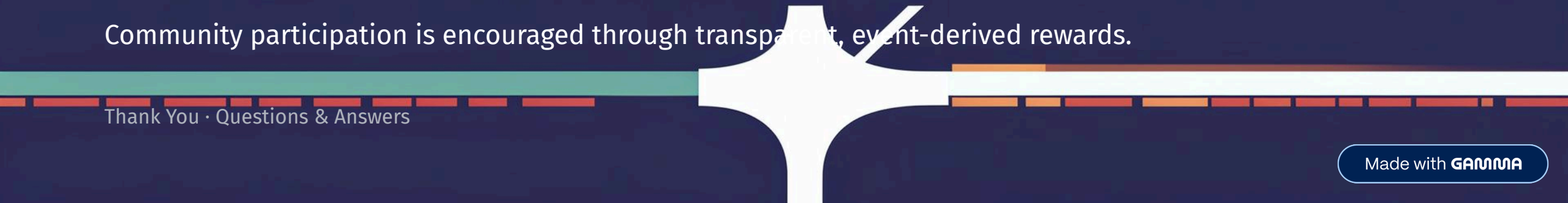
Smarter reporting. Fairer priorities. Better campuses.

CampusFix turns complaints into structured, ranked issues.

Gemma 4 handles semantic understanding.

Python retains control over safety, ranking, storage, and points.

Community participation is encouraged through transparent, event-derived rewards.



Thank You · Questions & Answers

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